Sashwat Tanay

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EMPLOYMENT

Assistant Professor Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru	2026-present
Postdoctoral Fellow LUX, Paris Observatory - PSL University	2023-2025
Adjunct Instructor University of Mississippi	2022-2023
Teaching and Research Assistant University of Mississippi	2016-2022
Junior Research Fellow Tata Institute of Fundamental Research, Mumbai	2013-2015

EDUCATION

Ph.D. (Physics) University of Mississippi (Advisor: Prof. Leo C. Stein)	2016-2022
B.Tech. (Mechanical Engineering) Indian Institute of Technology (IIT) Ropar	2009-2013

AWARDS & FELLOWSHIPS

PSL Postdoctoral Fellowship, Paris Observatory - PSL University	2023-2025
Graduate School Honors Fellowship, Univ. of Mississippi	2016-2020
Junior Research Fellowship, Tata Institute of Fundamental Research, Mumbai	2013-2015

RESEARCH INTERESTS

• Gravitational waves • Post-Newtonian dynamics of binary black holes • Quasi-normal mode ringdown of black holes • Hamiltonian dynamical systems • Extreme mass ratio inspirals

RESEARCH SUPERVISION

Tom Colin (master's student, Ecole Normale Supérieure, Paris) led to Publication (1)	Oct 2023-present
Manuel A. Morales (undergrad, Universidad Nacional de Trujillo)	Nov 2023-present
Rickmoy Samanta (postdoc, ISI Kolkata) led to Publication (5)	Sep 2021-Sep 2022
Pranav Kasetty (undergrad thesis co-advisor, IISc Bengaluru)	Oct 2021-Apr 2022

TEACHING EXPERIENCE

CC1105 Statistics and Probability I — Fall 2026
UTEC, Lima — course website & lecture notes

Phys 211 Physics for Science and Engineering I — Summer 2023
Univ. of Mississippi — course website & lecture notes

Phys 221, 222, 223, 224 Undergrad Physics Lab Courses (Teaching Assistant) — 2016–2022
Univ. of Mississippi

PROFESSIONAL SERVICE

Referee Physical Review & Physical Review Letters — Feb 2023-present

LANGUAGES

Fluent: English, Hindi **Elementary:** French (A1-A2)

INVITED TALKS & LECTURES

National University of Callao, Peru	Jan 2026
Pontifical Catholic University of Peru (Cafecitos Astronómicos)	Dec 2025
University of the Chinese Academy of Sciences	Sep 2025
Max Planck Institute for Gravitational Physics Potsdam (ACR Seminar)	Sep 2025
Max Planck Institute for Gravitational Physics Hannover (Gravitational Wave Group Meeting)	Jun 2025
University of Mississippi (Department Colloquium)	Apr 2025
IHES, Paris-Saclay University (Amplitudes and Gravitation Seminar-IHES/IPhT)	Jan 2025
York University, Toronto (Department Colloquium)	Dec 2024
Institut d'astrophysique de Paris, Sorbonne University (GReCO seminar)	Jan 2024
IISER Pune (Physics Seminar)	Jan 2024
Paris Observatory, Paris Sciences et Lettres (PSL) University (LUTH Seminar)	Sep 2023
Missouri University of Science and Technology (Department Colloquium)	Aug 2023
Northwestern University	Jul 2023
University of Illinois Urbana-Champaign (lecture workshop; lecture notes here)	Jun 2022
Montana State University (Relativity, Astrophysics and Space Science Seminar)	Apr 2022
Max Planck Institute for Gravitational Physics Potsdam (ACR Seminar)	Jun 2021
Simon Fraser University (Cosmology Seminar)	Sep 2020

OUTREACH & SERVICE

- Podcast with The Intergalactic Space Travellers Journal (Apple Podcasts, Spotify) — 2026
- Contributor on Wikipedia (English & French) including various articles on physics; see here and here
- Public Talk (Q2C, From Quantum to Cosmos), UTEC, Lima — 2025
- Public talk in French (Journée du LUTH), Paris Observatory — 2024
- Public Talk on Astronomy - Univ. of Mississippi — 2023
- Judge at The Speaker's Edge Competition - Univ. of Mississippi — 2022
- Organized STEM Summer Camp - Univ. of Mississippi — 2018, 2019
- Organized Spooky Physics Night - Univ. of Mississippi — 2016, 2017, 2018
- YouTube videos on research and popular science

PUBLICATIONS AND RESEARCH ARTICLES

1. T. Colin, **S. Tanay**, and L. Bernard. Analytical Solution of Spinning, Eccentric Binary Black Hole Dynamics at the Second Post-Newtonian Order, 2026, arXiv:2603.20031
2. V. Witzany, V. Skoupý, L. C. Stein, and **S. Tanay**. Actions of spinning compact binaries: Spinning particle in Kerr matched to dynamics at 1.5 post-Newtonian order, 2024, arXiv:2411.09742

3. **S. Tanay**. Towards a more robust algorithm for computing the Kerr quasinormal mode frequencies, 2022, arXiv:2210.03657

4. R. Samanta, **S. Tanay**, and L. C. Stein. Closed-form solutions of spinning, eccentric binary black holes at 1.5 post-Newtonian order. *Phys. Rev. D*, 108(14):124039, 2023, arXiv:2210.01605

5. **S. Tanay**, L. C. Stein, and G. Cho. Action-angle variables of a binary black hole with arbitrary eccentricity, spins, and masses at 1.5 post-Newtonian order. *Phys. Rev. D*, 107(26):103040, 2021, arXiv:2110.15351

6. G. Cho, **S. Tanay**, A. Gopakumar, and H. M. Lee. Generalized quasi-Keplerian solution for eccentric, nonspinning compact binaries at 4PN order and the associated inspiral-merger-ringdown waveform. *Phys. Rev. D*, 105(6):064010, 2022, arXiv: 2110.09608

7. **S. Tanay**, L. C. Stein, and J. T. Gálvez Ghersi. Integrability of eccentric, spinning black hole binaries up to second post-Newtonian order. *Phys. Rev. D*, 103(6):064066, 2021, arXiv: 2012.06586

8. **S. Tanay**, A. Klein, E. Berti, and A. Nishizawa. Convergence of Fourier-domain templates for inspiraling eccentric compact binaries. *Phys. Rev. D*, 100(6):064006, 2019, arXiv:1905.08811

9. **S. Tanay**, M. Haney, and A. Gopakumar. Frequency and time domain inspiral templates for comparable mass compact binaries in eccentric orbits. *Phys. Rev. D*, 93(6):064031, 2016, arXiv:1602.03081